

SegJSW: A Hybrid Model for Knee Osteoarthritis Classification

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Abstract—Knee osteoarthritis is currently one of the most serious and prevalent health issues in society. Over the years, numerous research have investigated applying Artificial Intelligence for early disease detection. However, most current approaches have merely relied on image processing techniques and thus not incorporating domain medical knowledge, which somehow limits the ultimate results. Therefore, in this paper, we propose a new method called *SegJSW*. Our model carries out the prediction process based on two branches of information: (1) overall diagnostic outcomes obtained from image classification models; and (2) Joint Space Width Evaluation, developed from our proposed method. Experimented on public dataset of actual patients, *SegJSW* yields significant accuracy improvements compared to the baseline method (81.09% vs. 77.65%), while also enjoying recall enhancement as well (77.38% vs. 68.73%).

Index Terms—hybrid model, segmentation, joint space width, classification, knee osteoarthritis.

I. INTRODUCTION

Knee osteoarthritis is one of modern society's most serious and common health issues. It is a type of joint inflammation where the protective cartilage in the knee joint becomes damaged and deteriorates, leading to direct contact between the bones within the joint. Without cartilage cushioning, the bones rub against each other, causing pain, swelling, and reduced joint mobility. Numerous studies have highlighted the prevalence of knee osteoarthritis worldwide. According

to a 2016 report by the *World Health Organization* (WHO), the prevalence of knee osteoarthritis among individuals aged 60 and above is 9.6% for men and 18.0% for women. Among those affected, 80% experience difficulties in mobility, and 25% face challenges in performing daily activities. These statistics demonstrate the significant impact of knee osteoarthritis on affected individuals' quality of life and self-care abilities.

In Vietnam, a study conducted by Nguyen *et al.* [1] in 2014 revealed that the prevalence of knee osteoarthritis is 31.2% for males and 35.3% for females. The research also indicated an increasing trend with age, with a prevalence rate of 8% in the 40-49 age group, 30% in the 50-59 age group, and a staggering 61.1% in the age group above 60. These figures are concerning, highlighting the rapid escalation of knee osteoarthritis with age and its significant impact on the Vietnamese population.

Researchers' interest in diagnosing knee osteoarthritis is growing. Various parameters from X-ray images are used to diagnose knee osteoarthritis, such as bone spurs and cartilage damage. Among them, *Joint Space Width* (JSW) has been proposed as one of the leading parameters to monitor the progression of knee osteoarthritis [3].

In addition to relying on the experience and expertise of physicians, the use of *Artificial Intelligence* (AI) as a diag-

nostic tool has achieved significant success in the healthcare field. There are two main approaches to applying AI in knee osteoarthritis diagnosis: *Machine Learning* (ML) [4], [5] and *Deep Learning* (DL) [6]–[9]. However, these methods have not yet achieved satisfactory and reliable results. This may be because these methods have not fully utilized diagnostic techniques from the medical field, particularly information regarding knee JSW. In particular, previous studies in Vietnam have not utilized this information. Therefore, we propose a new approach, known as *SegJSW*, that combines knee JSW information with overall diagnostic information obtained from certain image classification models (e.g., *CNN* or *ViT* [31]). In contrast to previous models that handle *X-ray* images, *SegJSW* framework focuses on measuring the degree of JSW and incorporates this information into the disease classification model. Our main contributions include:

- Propose a new method for evaluating knee JSW in *X-ray* images, based on the segmentation approach.
- Propose a new approach for diagnosing knee osteoarthritis based on two branches of information, which are overall diagnostic results from *X-ray* images and knee JSW.

The paper is organized into the following main sections: In Section II, we investigate the background of knee osteoarthritis and its correlation with knee JSW, as well as conduct a relevant literature review on the topic. Our proposed model will be presented in Section III. In Section IV, we provide experimental results and compare them with other models. In Section V, we will discuss this method. Finally, Section VI concludes our findings and discusses future work.

II. BACKGROUND AND RELATED WORKS

A. Knee Osteoarthritis and JSW

Knee osteoarthritis is a condition where the joint surfaces deteriorate, leading to the erosion of the joint cartilage and narrowing of the knee joint space. The friction between the exposed bone surfaces can cause further damage, destroying bone tissue. According to the Kellgren & Lawrence (KL) classification system [10], the levels of knee osteoarthritis are as follows (refer to Fig. 1).

- *Grade 0*: Healthy joint.
- *Grade 1*: Small osteophytes or suspected osteophytes.
- *Grade 2*: Clearly visible osteophytes, mild joint space narrowing.
- *Grade 3*: Obvious joint space narrowing, sclerosis beneath the cartilage, multiple and varying-sized osteophytes, possible bone deformity.
- *Grade 4*: Severe joint space narrowing (potentially complete loss of joint space), extensive sclerosis beneath the cartilage, large osteophytes, distinct bone deformity.

In medical practice, knee JSW is one of the crucial factors for the doctors to manually make their diagnosis on knee osteoarthritis levels. In Fig. 2, we illustrate a visual example of JSW which refers to the distance between the subchondral bone surfaces within the knee joint. It is a quantitative measure



Fig. 1: The severity of the Kellgren & Lawrence grading scale increases progressively from left to right.

often assessed using medical imaging methods such as *X-ray* or *MRI*. The distance between the subchondral bone surfaces reduces when the joint cartilage undergoes degeneration, as seen in knee osteoarthritis. This reduction in JSW is one of the most common signs of knee osteoarthritis.

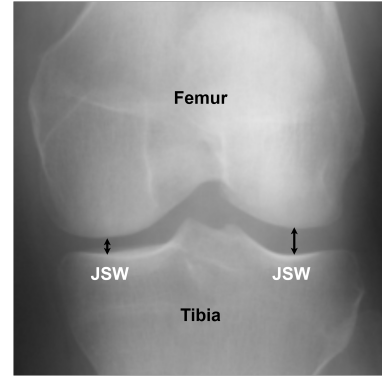


Fig. 2: Joint Space Width

B. Related Works

In recent years, there has been increasing interest in utilizing advanced medical imaging techniques for the assessment and classification of knee osteoarthritis. Numerous studies have indicated that the evaluation of the severity of knee osteoarthritis based on *X-ray* images can be improved through computer-aided methods [15]–[24]. Bone texture analysis is one of the primary approaches to automatically analyzing knee *X-ray* images [15]–[18]. Bone shape analysis of the bone is also another popular method for detecting osteoarthritis [17], [22]–[24]. Currently, advanced methods for automated detection and classification of knee osteoarthritis rely on deep learning techniques [19]–[21], particularly deep Convolutional Neural Networks (CNNs), which have achieved significant advancements.

However, these state-of-the-art deep learning models suffer from certain drawbacks. These approaches primarily rely on image processing but have not captured medical domain knowledge. To be precise, anatomical features known to be related to knee osteoarthritis are not taken into account. Information about knee JSW, which is a standard tool for evaluating the progression of knee osteoarthritis, is not addressed.

To accurately estimate JSW, image segmentation plays a crucial role. Segmentation for medical images can be divided into two directions: traditional segmentation and deep learning-based segmentation. Traditional segmentation methods cannot compare to deep learning methods in terms of

accuracy, but these ideas are still worth learning from [2], [6]–[8]. Some representative traditional segmentation methods include threshold-based segmentation [9], region-based segmentation [10], and edge detection-based segmentation [11]. These methods offer fast segmentation speeds but may not match the accuracy and level of detail achieved by deep learning methods.

Ambellan *et al.* [12] proposed a 2D Convolutional Neural Network for coarse segmentation and a 3D Convolutional Neural Network for fine segmentation, followed by image refinement based on statistics, which helped reduce noise from the background of knee images. Similarly, Gatti *et al.* [13] applied parallel 2D U-Net [33] and 3D U-Net [34] for coarse segmentation and an additional 3D U-Net [34] for fine segmentation. However, such architectures involving multiple sub-convolutional networks make the computations computationally intensive, and the input of the subsequent sub-convolutional networks is entirely dependent on the output of the previous sub-convolutional networks. Tan *et al.* [14] proposed an end-to-end collaborative multi-agent learning framework, but it is still limited by GPU memory constraints.

III. THE SEGJSW FRAMEWORK

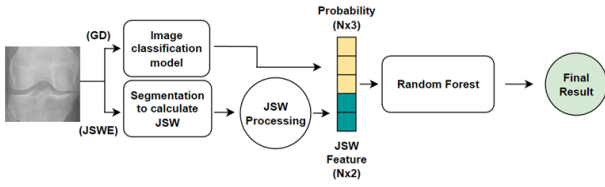


Fig. 3: Overview of SegJSW

Fig. 3 gives an overview of our proposed SegJSW framework. In this framework, our idea is to use certain image classification models (e.g., CNN or ViT [31]) to provide an overall diagnostic result from X-ray images and combine it with information about knee JSW to get the final result. To achieve this, our model consists of two branches: *GD* (General Diagnosis) and *JSWE* (JSW Evaluation). The role of the *GD* branch is to generate an overall diagnostic result from the X-ray images. Meanwhile, the *JSWE* branch is responsible to calculate the distance between the two components of the knee joint by segmenting the femur and tibia and then using a computational method to generate representation values for the degree of knee JSW. Finally, the information from both branches will be combined and passed through a Random Forest algorithm to get the result.

A. General diagnosis (GD)

In the training process, we trained the two branches of SegJSW independently. Fig. 4 illustrates the training process of the image classification model in the *GD* branch, where the model is trained to classify disease severity (the classification task). To continuously update the model parameters throughout the training process, we employed the Categorical Cross

Entropy Loss - $\mathcal{L}_{CE}$ as the loss function, which is defined in equation (1).

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{j=1}^N \sum_{i=1}^C Q_{ij} \log(P_{ij}) \quad (1)$$

where:

- P_{ij} and Q_{ij} correspondingly represent the probability distributions of the predicted outcome and the ground truth.
- C is the number of classes and N is the number of samples.

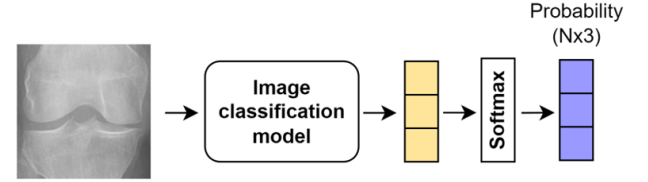


Fig. 4: Training process of the *GD* branch.

B. JSW Evaluation (JSWE)

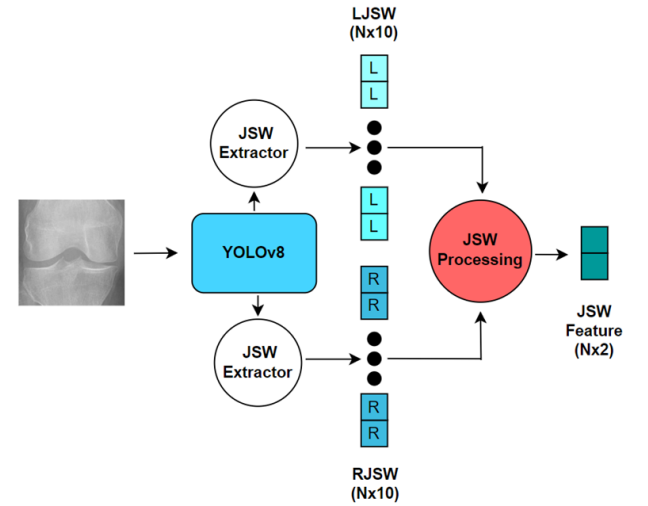


Fig. 5: Training process of the *JSWE* branch.

As mentioned earlier, JSW, the distance between the two bone surfaces in the knee joint, plays a crucial role in diagnosing knee osteoarthritis. JSW can aid in the early diagnosis of knee joint issues, support treatment decisions, and ensure effective monitoring of interventions. Therefore, we suggest evaluating the JSW from the original image and making full use of it for knee osteoarthritis diagnosis. Fig. 5 illustrates how we extract JSW features from X-ray images. In order to extract information about knee JSW for representation in a trainable model, we need two steps are *JSW Extractor* and *JSW Processing*.

JSW Extractor. In this step, to calculate JSW from processed and segmented knee joint images, we first identify the

contours of the femur and tibia from the segmented images. We select only the contours with the largest area for each bone. We limit the scope of the femur and tibia contours to ensure they are only within the knee joint region. An example is shown in Fig. 6. To calculate JSW, we need to remove the

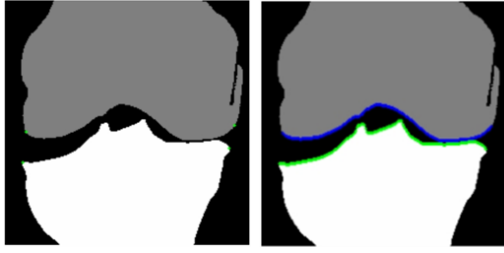


Fig. 6: On the left are the four boundary points, and on the right is the contour after being constrained.

central portion of the contour since JSW only considers the contact area between the two bone surfaces. The formula for determining the x -coordinate of the center point is calculated in equation (2).

$$p_x = \frac{x_{\min}^{\text{upper}} + x_{\max}^{\text{upper}} + x_{\min}^{\text{lower}} + x_{\max}^{\text{lower}}}{4} \quad (2)$$

where $x_{\min}^{\text{upper}}, x_{\max}^{\text{upper}}, x_{\min}^{\text{lower}}, x_{\max}^{\text{lower}}$ are respectively the minimum and maximum x -coordinates of the upper and lower contour. The points belonging to the contour with x -coordinates in the range $[p_x - r \times l / 2; p_x + r \times l / 2]$ will be removed, where:

$$l = \frac{x_{\max}^{\text{upper}} - x_{\min}^{\text{upper}} + x_{\max}^{\text{lower}} - x_{\min}^{\text{lower}}}{4} \quad (3)$$

and r is the removal rate (we chose $r = 0.2$).

After this step, the two contours are divided into the left upper contour, the right upper contour, the left lower contour, and the right lower contour. Since the positions of the upper and lower contours are not entirely aligned, we need to eliminate unnecessary discrepancies. Then, we calculate the corresponding distances between points with the same x -coordinates on the upper and lower contours, as illustrated in Fig. 7. The result is two distance vectors, one on the left side and one on the right side.

To reduce the impact of segmentation errors (where a few pixels may deviate abnormally) during the calculation, we project the two vectors into a 10-dimensional space. We divide each distance vector into ten approximately equal segments, with a maximum difference of 1. Then, each segment is aggregated using the mean function. After this step, we obtain two 10-dimensional vectors of $LJSW$ and $RJSW$, where $LJSW$ contains JSW information on the left side and $RJSW$ contains JSW information on the right side.

JSW Processing. From the previously obtained $LJSW$ and $RJSW$, we calculate two values: the average difference in narrowness between the two sides - JSW_{Mean} and the maximum difference between the two sides - JSW_{MM} , which are defined as follows.

$$JSW_{\text{Mean}} = \frac{|\text{mean}(LJSW) - \text{mean}(RJSW)|}{JSW_{\text{max}}} \quad (4)$$

$$JSW_{\text{MM}} = \frac{JSW_{\text{max}} - JSW_{\text{min}}}{JSW_{\text{max}}} \quad (5)$$

where:

- $JSW_{\text{max}} = \max(LJSW, RJSW)$
- $JSW_{\text{min}} = \begin{cases} \min(RJSW), & \text{if } JSW_{\text{max}} \in LJSW \\ \min(LJSW), & \text{otherwise} \end{cases}$
- $\text{mean}(X)$ is the average value of vector X

Both the JSW_{Mean} and JSW_{MM} values are divided by the maximum knee joint distance because the bone and cartilage structure can vary among individuals. By normalizing them with the individual's maximum distance value, we standardize them to a common value range.

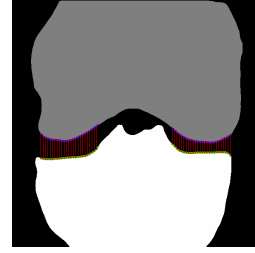


Fig. 7: Distances between corresponding points along the x -coordinate

C. Final classification with Random Forest

After completing the training process for both the GD and JSWE branches, we pass training dataset consisting of N images into the GD branch to obtain a sequence of vectors representing general diagnostic results $P = \{p_1, \dots, p_i, \dots, p_N\} \in R^{N \times C}$, and through the JSWE branch to obtain a sequence of vectors representing knee joint narrowing information $S = \{s_1, \dots, s_i, \dots, s_N\} \in R^{N \times 2}$. Next, we combine P and S to create $F = \{f_1, \dots, f_i, \dots, f_N\} \in R^{N \times (C+2)}$. Each vector in F contains information about the general diagnostic result and information about knee joint space width.

In this study, we train a Random Forest model on F to classify the severity of knee osteoarthritis. The goal is that when the general diagnostic result cannot clearly distinguish between two classes, the information about the JSW will help improve the accuracy of the final prediction results. The hyper parameters of the Random Forest model will be adjusted according to the image classification model we use in the GD branch.

IV. EXPERIMENT

A. Dataset

The X-ray image data used in our study was obtained from the *Osteoarthritis Initiative* (OAI) project, is publicly available at <https://nda.nih.gov/oai>. This dataset includes X-ray images of knee joints collected from 4796 volunteers, and each image was evaluated for the degree of osteoarthritis progression (according to the KL grade) by medical experts. In total, the dataset consists of 8260 labeled knee X-ray images.

Table I presents the distribution of the number of images in each part.

TABLE I: The distribution of the number of images corresponding to each class

Dataset	Class 0	Class 1	Class 2	Class 3	Class 4	Total
Train	2286	1046	1516	757	173	5778
Val	328	153	212	106	27	826
Test	639	296	447	223	51	1656
Total	3253	1495	2175	1086	251	8260

In the KL grading scale, class 1 is classified when there is suspicion of disease, which is highly challenging when relying solely on X-ray images. Doctors also need to consider clinical signs to make a diagnostic decision. This led us to come up with the action of re-labeling the data into three main classes as follows, as depicted in Table II.

- New class 0 (no disease): old class 0
- New class 1 (mild disease): old classes 1 and 2
- New class 2 (severe disease): old classes 3 and 4

TABLE II: Distribution of data among classes after re-labeling

Dataset	Class 0	Class 1	Class 2
Train	2286	2562	930
Val	328	365	133
Test	639	743	274
Total	3253	3670	1337

B. Experimental results

One of the major requirements when building classification models in the medical field is to minimize the model's failure to diagnose as few cases of disease as possible. Therefore, here we will utilize two evaluation methods simultaneously: *accuracy* and *recall*. Accuracy measures the overall correctness of the predictions, while recall measures the ability to correctly identify the number of disease cases within the entire dataset.

TABLE III: Compare the results of our method with the baseline method across multiple classification models

Model	Accuracy(%)		Recall(%)	
	Base	Our	Base	Our
DenseNet-121 [20]	81.21	81.21	76.79	78.36
ResNet-50 [25]	78.98	79.22	76.00	79.35
Inception-ResNet-V2 [26]	78.14	78.80	74.63	77.67
EfficientNet-V2-S [27]	77.65	81.09	68.73	77.38
ResNext-50 [28]	78.14	79.40	77.48	79.54
ConvNeXt-Base [29]	77.83	79.22	74.43	77.67
REGNETX-32GF [30]	80.25	79.52	81.41	82.10
ViT-B-16 [31]	76.21	77.11	69.71	72.76
Swin-B [32]	80.25	80.31	77.97	79.44

V. DISCUSSION

Based on the empirical results obtained (see Table III), we have observed a remarkable accuracy enhancement upon applying our proposed methodology for classification. This

unequivocally demonstrates the efficacy of integrating JSW information into the model evaluation process, thereby elevating proficiency in accurately categorizing disease instances. This improvement holds the potential to ascertain a more precise determination of the disease condition and facilitate the decision-making process about appropriate treatment strategies.

Another noteworthy aspect pertains to the substantial augmentation in recall exhibited by the models when our methodology is employed. This augmentation substantiates the method's proficiency in identifying positive cases, which is paramount in ensuring the comprehensive inclusion of disease-relevant instances without omissions. Fig. 8 provides specific evidence of the impact on performance metrics (accuracy and recall) when considering the presence or absence of JSW in decision-making. In addition, SegJSW still has some

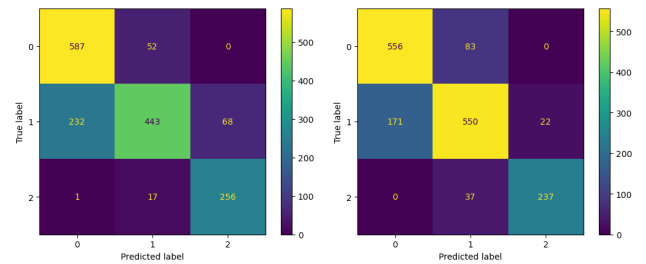


Fig. 8: Compare the confusion matrix between baseline method (left) vs. SegJSW (right) with EfficientNet-V2-Small [27] as image classification model.

limitations. Firstly, the influence of the overall diagnosis result on the final outcome is relatively high. If the overall diagnosis result is completely incorrect, the information about knee joint narrowing cannot support in providing a better result. Secondly, our model architecture is quite large, requiring a significant amount of resources and time for model training.

VI. CONCLUSION

In this paper, we introduced a new method for assessing knee JSW through segmentation and contour determination. Additionally, we reclassified the degree of osteoarthritis into three levels. We proposed an alternative approach for evaluating the severity of knee joint degeneration: combining the information of JSW from segmentation with the output of certain image classification model for the disease diagnosis. Our approach significantly improved diagnostic results when incorporating JSW compared to the baseline alone. These results highlighted the role of knee JSW in disease assessment. SegJSW exhibited high reliability by not only supporting disease diagnosis but also providing reasoning for the final decision. In the future, we will further develop this idea by incorporating additional types of information, such as osteophytes and subchondral bone thickness, for a comprehensive KL grading classification.

REFERENCES

- [1] Ho-Pham, L. T., Lai, T. Q., Mai, L. D., Doan, M. C., Pham, H. N., & Nguyen, T. V. (2014). Prevalence of radiographic osteoarthritis of the knee and its relationship to self-reported pain. *PloS one*, 9(4), e94563.
- [2] Gornale, S. S., Patravali, P. U., Marathe, K. S., & Hiremath, P. S. (2017). Determination of osteoarthritis using histogram of oriented gradients and multiclass SVM. *International Journal of Image, Graphics and Signal Processing*, 9(12), 41.
- [3] Tiulpin, A., Thevenot, J., Rahtu, E., Lehenkari, P., & Saarakkala, S. (2018). Automatic knee osteoarthritis diagnosis from plain radiographs: a deep learning-based approach. *Scientific reports*, 8(1), 1727.
- [4] Cruz-Roa, A. A., Arevalo Ovalle, J. E., Madabhushi, A., & González Osorio, F. A. (2013). A deep learning architecture for image representation, visual interpretability and automated basal-cell carcinoma cancer detection. In *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2013: 16th International Conference, Nagoya, Japan, September 22-26, 2013, Proceedings, Part II 16* (pp. 403-410). Springer Berlin Heidelberg.
- [5] Seise, M., McKenna, S. J., Ricketts, I. W., & Wigderowitz, C. A. (2005). Segmenting tibia and femur from knee X-ray images. In *Proc. of medical image understanding and analysis* (pp. 103-106).
- [6] Ma, Z., Tavares, J. M. R., & Jorge, R. N. (2009, February). A review on the current segmentation algorithms for medical images. In *International conference on imaging theory and applications* (Vol. 1, pp. 135-140). SciTePress.
- [7] Ferreira, A., Gentil, F., & Tavares, J. M. R. (2014). Segmentation algorithms for ear image data towards biomechanical studies. *Computer methods in biomechanics and biomedical engineering*, 17(8), 888-904.
- [8] Ma, Z., Tavares, J. M. R., Jorge, R. N., & Mascarenhas, T. (2010). A review of algorithms for medical image segmentation and their applications to the female pelvic cavity. *Computer Methods in Biomechanics and Biomedical Engineering*, 13(2), 235-246.
- [9] Xu, A., Wang, L., Feng, S., & Qu, Y. (2010, November). Threshold-based level set method of image segmentation. In *2010 Third International Conference on Intelligent Networks and Intelligent Systems* (pp. 703-706). IEEE.
- [10] Cigla, C., & Alatan, A. A. (2008, October). Region-based image segmentation via graph cuts. In *2008 15th IEEE International Conference on Image Processing* (pp. 2272-2275). IEEE.
- [11] Yu-Qian, Z., Wei-Hua, G., Zhen-Cheng, C., Jing-Tian, T., & Ling-Yun, L. (2006, January). Medical images edge detection based on mathematical morphology. In *2005 IEEE engineering in medicine and biology 27th annual conference* (pp. 6492-6495). IEEE.
- [12] Ambellan, F., Tack, A., Ehlike, M., & Zachow, S. (2019). Automated segmentation of knee bone and cartilage combining statistical shape knowledge and convolutional neural networks: Data from the Osteoarthritis Initiative. *Medical image analysis*, 52, 109-118.
- [13] Gatti, A. A., & Maly, M. R. (2021). Automatic knee cartilage and bone segmentation using multi-stage convolutional neural networks: data from the osteoarthritis initiative. *Magnetic Resonance Materials in Physics, Biology and Medicine*, 34(6), 859-875.
- [14] Tan, C., Yan, Z., Zhang, S., Li, K., & Metaxas, D. N. (2019). Collaborative multi-agent learning for MR knee articular cartilage segmentation. In *Medical Image Computing and Computer Assisted Intervention-MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13-17, 2019, Proceedings, Part II 22* (pp. 282-290). Springer International Publishing.
- [15] Lynch, J. A., Hawkes, D. J., & Buckland-Wright, J. C. (1991). Analysis of texture in macroradiographs of osteoarthritic knees, using the fractal signature. *Physics in Medicine & Biology*, 36(6), 709.
- [16] Hirvasniemi, J., Thevenot, J., Immonen, V., Liikavainio, T., Pulkkinen, P., Jämsä, T., ... & Saarakkala, S. (2014). Quantification of differences in bone texture from plain radiographs in knees with and without osteoarthritis. *Osteoarthritis and cartilage*, 22(10), 1724-1731.
- [17] Thomson, J., O'Neill, T., Felson, D., & Cootes, T. (2015). Automated shape and texture analysis for detection of osteoarthritis from radiographs of the knee. In *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part II 18* (pp. 127-134). Springer International Publishing.
- [18] Janvier, T., Jennane, R., Toumi, H., & Lespessailles, E. (2017). Subchondral tibial bone texture predicts the incidence of radiographic knee osteoarthritis: data from the Osteoarthritis Initiative. *Osteoarthritis and cartilage*, 25(12), 2047-2054.
- [19] Antony, J., McGuinness, K., O'Connor, N. E., & Moran, K. (2016, December). Quantifying radiographic knee osteoarthritis severity using deep convolutional neural networks. In *2016 23rd international conference on pattern recognition (ICPR)* (pp. 1195-1200). IEEE.
- [20] Norman, B., Pedoia, V., Noworolski, A., Link, T. M., & Majumdar, S. (2019). Applying densely connected convolutional neural networks for staging osteoarthritis severity from plain radiographs. *Journal of digital imaging*, 32(3), 471-477.
- [21] Tiulpin, A., Thevenot, J., Rahtu, E., Lehenkari, P., & Saarakkala, S. (2018). Automatic knee osteoarthritis diagnosis from plain radiographs: a deep learning-based approach. *Scientific reports*, 8(1), 1727.
- [22] Minciullo, L., & Cootes, T. (2016, December). Fully automated shape analysis for detection of Osteoarthritis from lateral knee radiographs. In *2016 23rd international conference on pattern recognition (ICPR)* (pp. 3787-3791). IEEE.
- [23] Haverkamp, D. J., Schiphof, D., Bierma-Zeinstra, S. M., Weinans, H., & Waarsing, J. H. (2011). Variation in joint shape of osteoarthritic knees. *Arthritis & Rheumatism*, 63(11), 3401-3407.
- [24] Martinez, A. M., Flament, I., Liu, F., Lee, J., Cao, P., Majumdar, S., & Pedoia, V. (2019). Discovering knee osteoarthritis bone shape features using deep learning. *Osteoarthritis and Cartilage*, 27, S386-S387.
- [25] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).
- [26] Szegedy, C., Ioffe, S., Vanhoucke, V., & Alemi, A. (2017, February). Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 31, No. 1).
- [27] Tan, M., & Le, Q. (2021, July). Efficientnetv2: Smaller models and faster training. In *International conference on machine learning* (pp. 10096-10106). PMLR.
- [28] Xie, S., Girshick, R., Dollár, P., Tu, Z., & He, K. (2017). Aggregated residual transformations for deep neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1492-1500).
- [29] Liu, Z., Mao, H., Wu, C. Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2020). A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 11976-11986).
- [30] Radosavovic, I., Kosaraju, R. P., Girshick, R., He, K., & Dollár, P. (2020). Designing network design spaces. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10428-10436).
- [31] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.
- [32] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., ... & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 10012-10022).
- [33] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18* (pp. 234-241). Springer International Publishing.
- [34] Çiçek, Ö., Abdulkadir, A., Lienkamp, S. S., Brox, T., & Ronneberger, O. (2016). 3D U-Net: learning dense volumetric segmentation from sparse annotation. In *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2016: 19th International Conference, Athens, Greece, October 17-21, 2016, Proceedings, Part II 19* (pp. 424-432). Springer International Publishing.